LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: The Fundamental Unit of Life (Cell)

Chapter / Topic: The Cell (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Draw neat, labelled diagrams wherever required.

SECTION A

6 marks

Choose the correct option.

Q1. The nucleus of the cell was discovered by: **[1]**

(a) Robert Hooke (b) Leeuwenhoek
(c) Purkinje (d) Robert Brown

Q2. Which organelle is called the 'kitchen of the cell' in plants? **[1]**

(a) Mitochondria (b) Ribosome
(c) Chloroplast (d) Lysosome

Q3. Which of these is absent in animal cells? **[1]**

(a) Cell membrane (b) Cell wall
(c) Nucleus (d) Mitochondria

Q4. The movement of water molecules through a selectively permeable membrane is called: **[1]**

(a) osmosis (b) diffusion
(c) endocytosis (d) plasmolysis

Q5. Which organelle is involved in forming lysosomes? **[1]**

(a) Ribosome (b) Golgi apparatus
(c) Nucleus (d) Plastid

Q6. **Assertion (A):** Mitochondria are able to make some of their own proteins.
Reason (R): Mitochondria have their own DNA and ribosomes. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why is the plasma membrane called a selectively permeable membrane? **[2]**

OR

What is endocytosis? Give one example.

Q8. Name two organelles that have their own genetic material (DNA). **[2]**

SECTION C

6 marks

Q9. Name the cell part that: **[3]**

(i) helps detoxify poisons and drugs in liver cells. **[1]**

(ii) makes the proteins and lipids needed to build the cell membrane (membrane biogenesis). **[1]**

(iii) stores starch, oils and proteins in plant cells. **[1]**

Q10. Give three differences between plant cells and animal cells. **[3]**

OR

What happens to a cell placed in (a) a hypotonic solution, (b) a hypertonic solution, (c) an isotonic solution?

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Describe the structure and functions of the nucleus. **[3]**

(ii) Name the two main types of cell division and give the purpose of each. **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A student removed the shell of an egg by dissolving it in dilute hydrochloric acid, leaving the egg covered only by its thin membrane. The egg was put in pure water, and after 5 minutes it had swollen. A similar egg was put in a concentrated salt solution, and it shrank. **[4]**

(i) Why did the egg swell in pure water? **[1]**

(ii) Why did the egg shrink in the salt solution? **[1]**

(iii) Name the process involved and the property of the membrane that makes it possible. Give one everyday example of the same process. **[2]**

~ All the best! ~